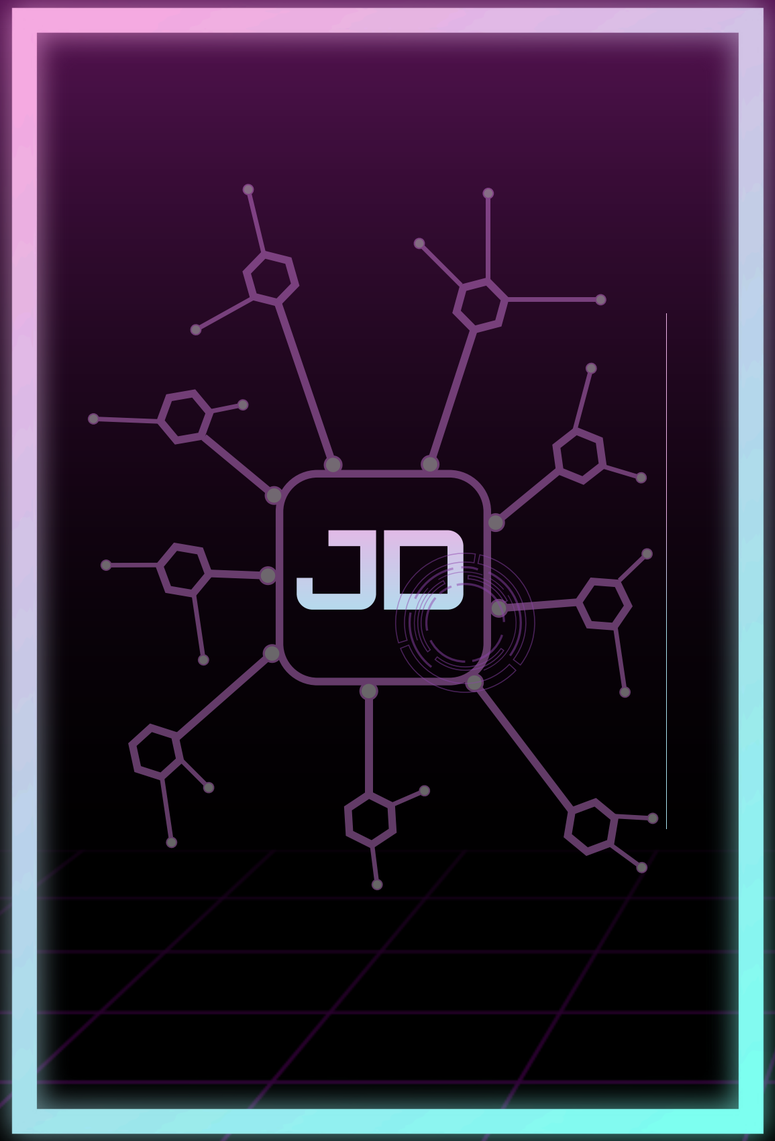



DAYSH UDDHARAKAR_LTD

# JD_LTD COMPANY

## DATA BASE ARCHITECTURE & SYSTEMS REPORT

**Organization:** JD_ltd company
**Project Scope:** E-Commerce Supply Chain Optimization
**Stack:** MySQL Server 8.0 / Schema Design & Optimization

# ABOUT US →

**JD_Ltd** is a leading modern technology provider and workplace infrastructure supplier that specializes in equipping businesses, developers, and creators with high-performance computing hardware, premium peripherals, professional audio/video gear, and ergonomic workstation setups. As an industry-focused enterprise, the company masterfully handles a **vast , fast-moving technical inventory** alongside high-volume supply chain operations, bridging the gap between real-time product availability and rapid order fulfillment through synchronized, secure data pipelines. By maintaining absolute data integrity and tracking every stock variation seamlessly, JD_Ltd eliminates traditional procurement bottlenecks such as backorders or unexpected stockouts, creating a **friction-free experience for corporate clients.** Enterprise teams can effortlessly execute complex, multi-item bulk procurement orders—outfitting entire offices or specialized development labs in a single transaction—while benefiting from **transparent, data-driven pricing models** that allow for predictable infrastructure forecasting with zero hidden costs or checkout spikes.

# PRIMARY AIM →

The primary aim of **JD_Ltd** is to seamlessly equip businesses, developers, and creators with high-performance computing hardware and optimized workplace infrastructure. To achieve this, the company focuses on three core objectives:

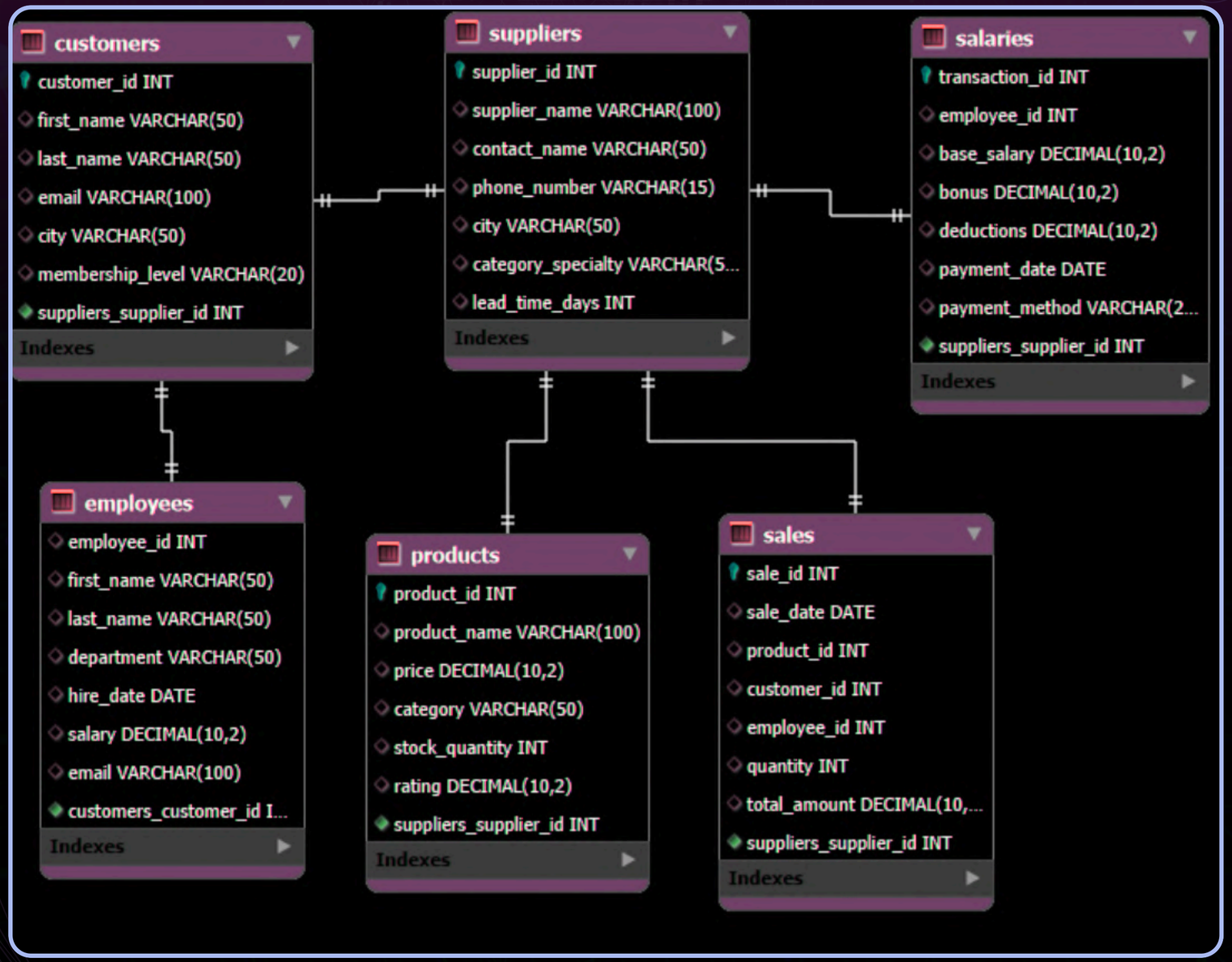
- **Eliminate Procurement Bottlenecks:** The company aims to completely remove traditional delays, backorders, and unexpected stockouts by maintaining real-time, synchronized data pipelines across its entire technical inventory.
- **Provide a Friction-Free Client Experience:** A major goal is to make bulk procurement simple, allowing corporate clients to outfit entire offices or development labs in a single, unified transaction.
- **Ensure Pricing Transparency:** JD_Ltd aims to deliver predictable infrastructure forecasting for its clients by utilizing data-driven pricing models that completely eliminate hidden fees and sudden price spikes.

# PROJECT OBJECTIVES →

The **project objectives** bridge the gap between the company's business needs and technical execution. The primary goals of this database architecture and analytics project for **JD_Ltd** include:

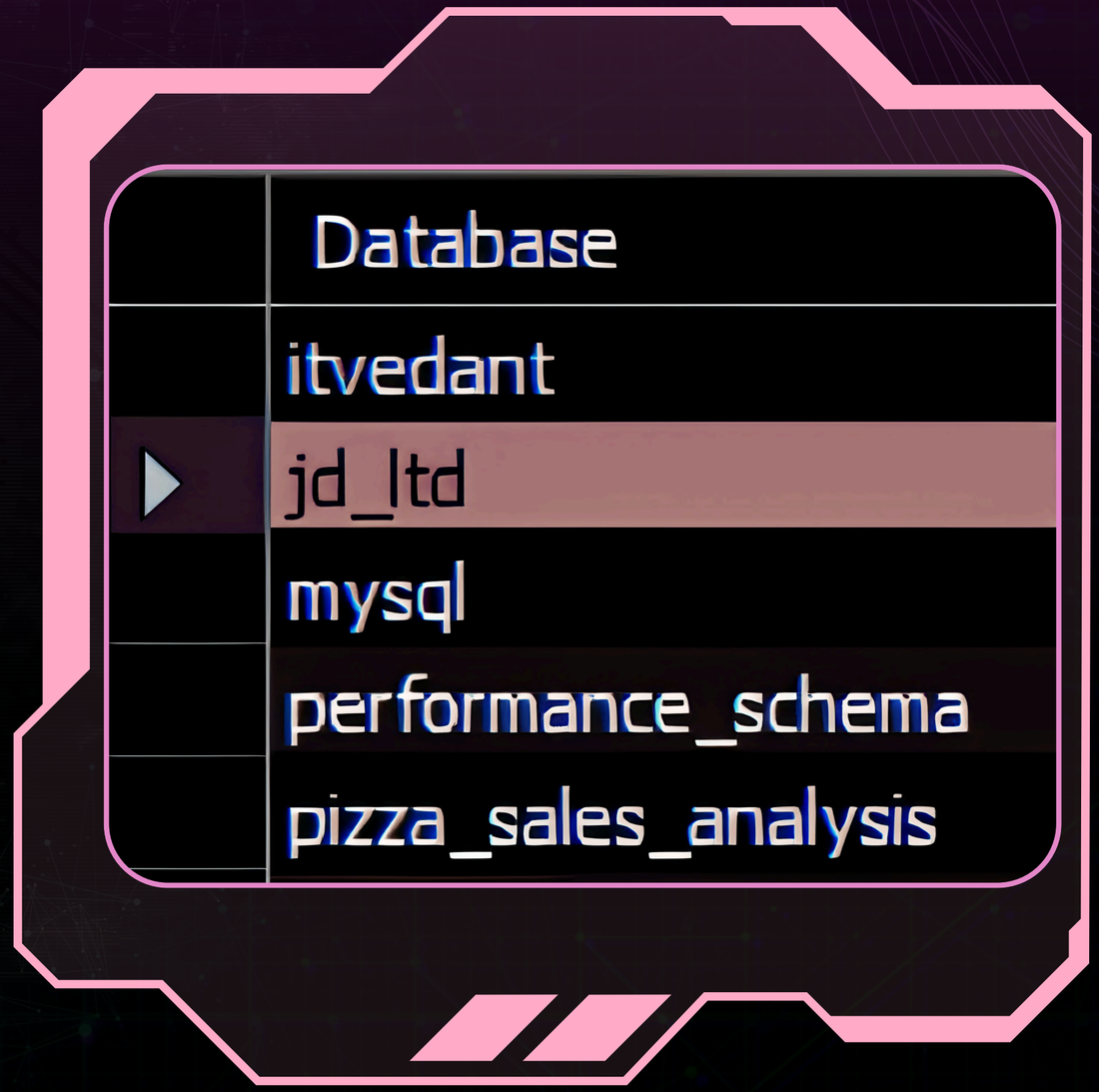
- **Design a Scalable Relational Architecture:** Build and normalize a secure backend database schema (products and sales) capable of tracking a vast, fast-moving technical inventory and high-frequency transactional data simultaneously.
- **Eliminate Data Blind Spots:** Ensure flawless structural synchronization between the client catalog and core order ledgers to prevent data defects, allowing the company to track real-time product availability accurately and eliminate procurement bottlenecks.
- **Extract Advanced Business Intelligence:** Utilize complex SQL aggregations, LEFT JOIN operations, and nested subqueries to compute key performance metrics—such as isolating true unit prices after promotional adjustments and identifying active customer lifecycle patterns.
- **Enforce Strict Production Safeguards:** Establish secure operational maintenance protocols by properly navigating database constraints and safety configurations (like toggling SQL_SAFE_UPDATES), ensuring broad-scope overwrites or table resets can be executed without disrupting live pipelines.

# ER DIAGRAM


customers
customer_id INT
first_name VARCHAR(50)
last_name VARCHAR(50)
email VARCHAR(100)
city VARCHAR(50)
membership_level VARCHAR(20)
suppliers_supplier_id INT
Indexes
suppliers
supplier_id INT
supplier_name VARCHAR(100)
contact_name VARCHAR(50)
phone_number VARCHAR(15)
city VARCHAR(50)
category_specialty VARCHAR(5...
lead_time_days INT
Indexes
salaries
transaction_id INT
employee_id INT
base_salary DECIMAL(10,2)
bonus DECIMAL(10,2)
deductions DECIMAL(10,2)
payment_date DATE
payment_method VARCHAR(2...
suppliers_supplier_id INT
Indexes
employees
employee_id INT
first_name VARCHAR(50)
last_name VARCHAR(50)
department VARCHAR(50)
hire_date DATE
salary DECIMAL(10,2)
email VARCHAR(100)
customers_customer_id I...
Indexes
products
product_id INT
product_name VARCHAR(100)
price DECIMAL(10,2)
category VARCHAR(50)
stock_quantity INT
rating DECIMAL(10,2)
suppliers_supplier_id INT
Indexes
sales
sale_id INT
sale_date DATE
product_id INT
customer_id INT
employee_id INT
quantity INT
total_amount DECIMAL(10,...
suppliers_supplier_id INT
Indexes

# CREATING DATABASE :

```
create database
JD_ltd;
use JD_ltd;
SHOW DATABASES;
```



# TABLES IN JD_LTD

```
show tables;
```



| Tables_in_jd_ltd |
| --- |
| customers |
| employees |
| products |
| salaries |
| sales |
| suppliers |

# CREATING TABLES :

## 01 Products

```
create table products(
product_id int primary key,
product_name varchar(100),
price decimal(10,2),
category varchar(50),
stock_quantity int ,
rating decimal(10,2)
);
```



| | Field | Type | Null | Key |
|---|---|---|---|---|
| ▶ | product_id | int | NO | PRI |
| | product_name | varchar(100) | YES | |
| | price | decimal(10,2) | YES | |
| | category | varchar(50) | YES | |
| | stock_quantity | int | YES | |
| | rating | decimal(10,2) | YES | |

# CREATING TABLES :

## 02 Suppliers

```
CREATE TABLE suppliers (
supplier_id INT PRIMARY KEY,
supplier_name VARCHAR(100),
contact_name VARCHAR(50),
phone_number VARCHAR(15),
city VARCHAR(50),
category_specialty VARCHAR(50),
lead_time_days INT
);
```

| Field | Type | Null | Key |
|---|---|---|---|
| supplier_id | int | NO | PRI |
| supplier_name | varchar(100) | YES | |
| contact_name | varchar(50) | YES | |
| phone_number | varchar(15) | YES | |
| city | varchar(50) | YES | |
| category_specialty | varchar(50) | YES | |
| lead_time_days | int | YES | |

# CREATING TABLES :

## 03 Employees

```
CREATE TABLE employees (
employee_id INT PRIMARY KEY,
first_name VARCHAR(50),
last_name VARCHAR(50),
department VARCHAR(50),
hire_date DATE,
salary DECIMAL(10, 2),
email VARCHAR(100)
);
```



| Field | Type | Null | Key |
|---|---|---|---|
| employee_id | int | NO | PRI |
| first_name | varchar(50) | YES | |
| last_name | varchar(50) | YES | |
| department | varchar(50) | YES | |
| hire_date | date | YES | |
| salary | decimal(10,2) | YES | |
| email | varchar(100) | YES | |

# CREATING TABLES :

## 04 Salaries

```
CREATE TABLE salaries (
transaction_id INT PRIMARY KEY,
employee_id INT,
base_salary DECIMAL(10, 2),
bonus DECIMAL(10, 2),
deductions DECIMAL(10, 2),
payment_date DATE,
payment_method VARCHAR(20)
);
```



| Field | Type | Null | Key |
|---|---|---|---|
| transaction_id | int | NO | PRI |
| employee_id | int | YES | |
| base_salary | decimal(10,2) | YES | |
| bonus | decimal(10,2) | YES | |
| deductions | decimal(10,2) | YES | |
| payment_date | date | YES | |
| payment_method | varchar(20) | YES | |

# CREATING TABLES :

## 05 Sales

```
CREATE TABLE sales (
sale_id INT PRIMARY KEY,
sale_date DATE,
product_id INT,
customer_id INT,
employee_id INT,
quantity INT,
total_amount DECIMAL(10, 2)
);
```



| | Field | Type | Null | Key |
|---|---|---|---|---|
| ▶ | sale_id | int | NO | PRI |
| | sale_date | date | YES | |
| | product_id | int | YES | |
| | customer_id | int | YES | |
| | employee_id | int | YES | |
| | quantity | int | YES | |
| | total_amount | decimal(10,2) | YES | |

# CREATING TABLES :

## 06 Customers

```
CREATE TABLE customers (
customer_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
        email VARCHAR(100),
        city VARCHAR(50),
        membership_level
        VARCHAR(20)  );
```



| | Field | Type | Null | Key |
|---|---|---|---|---|
| ▶ | customer_id | int | NO | PRI |
| | first_name | varchar(50) | YES | |
| | last_name | varchar(50) | YES | |
| | email | varchar(100) | YES | |
| | city | varchar(50) | YES | |
| | membership_level | varchar(20) | YES | |

# INSERTING VALUES INTO TABLES (PRODUCTS) :

```
value(101, 'Mechanical Keyboard K9', 350.50, 'Peripherals', 45, 4.8),
(102, 'Ergonomic Mouse M2', 550.00, 'Peripherals', 120, 4.5),
(103, 'Ultra-Wide Monitor 34', 45000.00, 'Display', 15, 4.9),
(104, 'Noise Cancelling Headphones', 1999.99, 'Audio', 30, 4.7),
(105, 'USB-C Hub 7-in-1', 450.00, 'Accessories', 200, 4.2),
(106, 'Wireless Charging Pad', 299.99, 'Accessories', 85, 4.0),
(107, 'Smart LED Desk Lamp', 899.00, 'Office', 60, 4.6),
(108, 'Portable SSD 1TB', 4110.00, 'Storage', 50, 4.8),
(109, 'HD Webcam 1080p', 750.00, 'Audio/Video', 40, 4.3),
(110, 'Aluminum Laptop Stand', 850.50, 'Office', 110, 4.4),
(111, 'Bluetooth Speaker V4', 799.99, 'Audio', 75, 4.2),
(112, '4K Mirrorless Camera', 2999.00, 'Photography', 12, 4.9),
(113, 'Gaming Chair Pro', 4500.00, 'Furniture', 25, 4.6),
(114, 'Smartwatch Series 5', 1999.50, 'Wearables', 60, 4.4),
(115, 'External Hard Drive 2TB', 2599.00, 'Storage', 45, 4.5),
(116, 'Standing Desk Frame', 929.00, 'Furniture', 18, 4.7),
(117, 'Studio Microphone', 549.00, 'Audio', 33, 4.8),
(118, 'VR Headset Gen 2', 6999.00, 'Gaming', 10, 4.3),
(119, 'Mechanical Keycaps Set', 350.00, 'Accessories', 150, 4.1),
(120, 'Vertical Laptop Dock', 499.99, 'Office', 90, 4.5),
(121, 'Thunderbolt 4 Cable', 250.00, 'Accessories', 300, 4.0),
(122, 'Wireless Lapel Mic', 899.00, 'Audio', 55, 4.6),
(123, 'Curved Gaming Monitor', 29999.99, 'Display', 22, 4.9),
(124, 'Anti-Static Work Mat', 1299.50, 'Tools', 100, 4.2),
(125, 'Precision Tool Kit', 699.00, 'Tools', 40, 4.7);
```

| product_id | product_name | price | category | stock_quantity | rating |
|---|---|---|---|---|---|
| 101 | Mechanical Keyboard K9 | 350.50 | Peripherals | 45 | 4.80 |
| 102 | Ergonomic Mouse M2 | 550.00 | Peripherals | 120 | 4.50 |
| 103 | Ultra-Wide Monitor 34 | 45000.00 | Display | 15 | 4.90 |
| 104 | Noise Cancelling Headphones | 1999.99 | Audio | 30 | 4.70 |
| 105 | USB-C Hub 7-in-1 | 450.00 | Accessories | 200 | 4.20 |
| 106 | Wireless Charging Pad | 299.99 | Accessories | 85 | 4.00 |
| 107 | Smart LED Desk Lamp | 899.00 | Office | 60 | 4.60 |
| 108 | Portable SSD 1TB | 4110.00 | Storage | 50 | 4.80 |
| 109 | HD Webcam 1080p | 750.00 | Audio/Video | 40 | 4.30 |
| 110 | Aluminum Laptop Stand | 850.50 | Office | 110 | 4.40 |
| 111 | Bluetooth Speaker V4 | 799.99 | Audio | 75 | 4.20 |
| 112 | 4K Mirrorless Camera | 2999.00 | Photography | 12 | 4.90 |
| 113 | Gaming Chair Pro | 4500.00 | Furniture | 25 | 4.60 |
| 114 | Smartwatch Series 5 | 1999.50 | Wearables | 60 | 4.40 |
| 115 | External Hard Drive 2TB | 2599.00 | Storage | 45 | 4.50 |
| 116 | Standing Desk Frame | 929.00 | Furniture | 18 | 4.70 |
| 117 | Studio Microphone | 549.00 | Audio | 33 | 4.80 |
| 118 | VR Headset Gen 2 | 6999.00 | Gaming | 10 | 4.30 |
| 119 | Mechanical Keycaps Set | 350.00 | Accessories | 150 | 4.10 |
| 120 | Vertical Laptop Dock | 499.99 | Office | 90 | 4.50 |
| 121 | Thunderbolt 4 Cable | 250.00 | Accessories | 300 | 4.00 |
| 122 | Wireless Lapel Mic | 899.00 | Audio | 55 | 4.60 |
| 123 | Curved Gaming Monitor | 29999.99 | Display | 22 | 4.90 |
| 124 | Anti-Static Work Mat | 1299.50 | Tools | 100 | 4.20 |
| 125 | Precision Tool Kit | 699.00 | Tools | 40 | 4.70 |
| NULL | NULL | NULL | NULL | NULL | NULL |

# • INSERTING VALUES INTO TABLES (SUPPLIERS) :

```
(201, 'Global Tech Imports', 'John Doe', '9876543210', 'Mumbai',
'Peripherals', 14),
(202, 'Indo-Gear Manufacturing', 'Rajesh Kumar', '9822334455',
'Pune', 'Furniture', 7),
(203, 'Silicon Source', 'Sarah Lee', '9123456789', 'Bangalore',
'Storage', 10),
(204, 'Audio Pro Wholesalers', 'Mike Ross', '9988776655', 'Delhi',
'Audio', 5),
(205, 'Bright Light Co.', 'Anita Birla', '9766554433', 'Chennai',
'Office', 3),
(206, 'Precision Tools Ltd', 'Sanjay Dutt', '9555444333',
'Ahmedabad', 'Tools', 6),
(207, 'Visionary Displays', 'Ellen Park', '9444333222',
'Hyderabad', 'Display', 12),
(208, 'SmartWear Solutions', 'Kevin Vats', '9333222111', 'Gurgaon',
'Wearables', 8),
(209, 'Eco-Packaging Inc', 'Meera Iyer', '9222111000', 'Mumbai',
'Accessories', 4),
(210, 'Alpha Photography', 'Robert Sen', '9111000999', 'Kolkata',
'Photography', 15);
```

| supplier_id | supplier_name | contact_name | phone_number | city | category_specialty | lead_time_days |
|---|---|---|---|---|---|---|
| 201 | Global Tech Imports | John Doe | 9876543210 | Mumbai | Peripherals | 14 |
| 202 | Indo-Gear Manufacturing | Rajesh Kumar | 9822334455 | Pune | Furniture | 7 |
| 203 | Silicon Source | Sarah Lee | 9123456789 | Bangalore | Storage | 10 |
| 204 | Audio Pro Wholesalers | Mike Ross | 9988776655 | Delhi | Audio | 5 |
| 205 | Bright Light Co. | Anita Birla | 9766554433 | Chennai | Office | 3 |
| 206 | Precision Tools Ltd | Sanjay Dutt | 9555444333 | Ahmedabad | Tools | 6 |
| 207 | Visionary Displays | Ellen Park | 9444333222 | Hyderabad | Display | 12 |
| 208 | SmartWear Solutions | Kevin Vats | 9333222111 | Gurgaon | Wearables | 8 |
| 209 | Eco-Packaging Inc | Meera Iyer | 9222111000 | Mumbai | Accessories | 4 |
| 210 | Alpha Photography | Robert Sen | 9111000999 | Kolkata | Photography | 15 |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL |

# • INSERTING VALUES INTO TABLES (EMPLOYEES) :

```
(1, 'Aravind', 'Kumar', 'Sales', '2023-01-15', 45000.00,
'aravind.k@quantumgear.com'),
(2, 'Sneha', 'Patil', 'Marketing', '2023-03-20', 48000.00,
'sneha.p@quantumgear.com'),
(3, 'Rohan', 'Sharma', 'IT Support', '2022-11-10', 55000.00,
'rohan.s@quantumgear.com'),
(4, 'Priya', 'Iyer', 'Sales', '2024-02-01', 42000.00,
'priya.i@quantumgear.com'),
(5, 'Vikram', 'Singh', 'Logistics', '2023-06-12', 38000.00,
'vikram.s@quantumgear.com'),
(6, 'Ananya', 'Reddy', 'HR', '2022-08-25', 52000.00,
'ananya.r@quantumgear.com'),
(7, 'Kabir', 'Mehta', 'Finance', '2021-12-05', 65000.00,
'kabir.m@quantumgear.com'),
(8, 'Ishita', 'Desai', 'Marketing', '2024-01-10', 47000.00,
'ishita.d@quantumgear.com'),
(9, 'Arjun', 'Verma', 'IT Support', '2023-09-18', 54000.00,
'arjun.v@quantumgear.com'),
(10, 'Siddharth', 'Joshi', 'Sales', '2023-04-30', 44000.00,
'sid.j@quantumgear.com'),
(11, 'Meera', 'Nair', 'Finance', '2022-02-14', 63000.00,
'meera.n@quantumgear.com'),
(12, 'Aditya', 'Gupta', 'Logistics', '2023-11-22', 39000.00,
'aditya.g@quantumgear.com'),
(13, 'Tanvi', 'Bhasin', 'HR', '2023-07-05', 51000.00,
'tanvi.b@quantumgear.com'),
(14, 'Rishi', 'Kapoor', 'Marketing', '2024-03-15', 46000.00,
'rishi.k@quantumgear.com'),
(15, 'Sana', 'Khan', 'Sales', '2022-10-01', 45500.00,
'sana.k@quantumgear.com');
```

| employee_id | first_name | last_name | department | hire_date | salary | email |
|---|---|---|---|---|---|---|
| 1 | Aravind | Kumar | Sales | 2023-01-15 | 45000.00 | aravind.k@quantumgear.com |
| 2 | Sneha | Patil | Marketing | 2023-03-20 | 48000.00 | sneha.p@quantumgear.com |
| 3 | Rohan | Sharma | IT Support | 2022-11-10 | 55000.00 | rohan.s@quantumgear.com |
| 4 | Priya | Iyer | Sales | 2024-02-01 | 42000.00 | priya.i@quantumgear.com |
| 5 | Vikram | Singh | Logistics | 2023-06-12 | 38000.00 | vikram.s@quantumgear.com |
| 6 | Ananya | Reddy | HR | 2022-08-25 | 52000.00 | ananya.r@quantumgear.com |
| 7 | Kabir | Mehta | Finance | 2021-12-05 | 65000.00 | kabir.m@quantumgear.com |
| 8 | Ishita | Desai | Marketing | 2024-01-10 | 47000.00 | ishita.d@quantumgear.com |
| 9 | Arjun | Verma | IT Support | 2023-09-18 | 54000.00 | arjun.v@quantumgear.com |
| 10 | Siddharth | Joshi | Sales | 2023-04-30 | 44000.00 | sid.j@quantumgear.com |
| 11 | Meera | Nair | Finance | 2022-02-14 | 63000.00 | meera.n@quantumgear.com |
| 12 | Aditya | Gupta | Logistics | 2023-11-22 | 39000.00 | aditya.g@quantumgear.com |
| 13 | Tanvi | Bhasin | HR | 2023-07-05 | 51000.00 | tanvi.b@quantumgear.com |
| 14 | Rishi | Kapoor | Marketing | 2024-03-15 | 46000.00 | rishi.k@quantumgear.com |
| 15 | Sana | Khan | Sales | 2022-10-01 | 45500.00 | sana.k@quantumgear.com |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL |

# • INSERTING VALUES INTO TABLES (SALARIES) :

```
(501, 1, 45000.00, 2000.00, 1500.00, '2026-04-30', 'Bank Transfer'),
(502, 2, 48000.00, 1500.00, 1200.00, '2026-04-30', 'Bank Transfer'),
(503, 3, 55000.00, 3000.00, 2000.00, '2026-04-30', 'Check'),
(504, 4, 42000.00, 0.00, 1000.00, '2026-04-30', 'Bank Transfer'),
(505, 5, 38000.00, 500.00, 800.00, '2026-04-30', 'Bank Transfer'),
(506, 6, 52000.00, 2500.00, 1800.00, '2026-04-30', 'Bank Transfer'),
(507, 7, 65000.00, 5000.00, 3000.00, '2026-04-30', 'Check'),
(508, 8, 47000.00, 1200.00, 1100.00, '2026-04-30', 'Bank Transfer'),
(509, 9, 54000.00, 2200.00, 1600.00, '2026-04-30', 'Bank Transfer'),
(510, 10, 44000.00, 1000.00, 1000.00, '2026-04-30', 'Bank Transfer'),
(511, 11, 63000.00, 4000.00, 2500.00, '2026-04-30', 'Check'),
(512, 12, 39000.00, 600.00, 900.00, '2026-04-30', 'Bank Transfer'),
(513, 13, 51000.00, 2000.00, 1500.00, '2026-04-30', 'Bank Transfer'),
(514, 14, 46000.00, 1100.00, 1100.00, '2026-04-30', 'Bank Transfer'),
(515, 15, 45500.00, 1800.00, 1400.00, '2026-04-30', 'Bank Transfer');
```

| transaction_id | employee_id | base_salary | bonus | deductions | payment_date | payment_method |
|---|---|---|---|---|---|---|
| 501 | 1 | 45000.00 | 2000.00 | 1500.00 | 2026-04-30 | Bank Transfer |
| 502 | 2 | 48000.00 | 1500.00 | 1200.00 | 2026-04-30 | Bank Transfer |
| 503 | 3 | 55000.00 | 3000.00 | 2000.00 | 2026-04-30 | Check |
| 504 | 4 | 42000.00 | 0.00 | 1000.00 | 2026-04-30 | Bank Transfer |
| 505 | 5 | 38000.00 | 500.00 | 800.00 | 2026-04-30 | Bank Transfer |
| 506 | 6 | 52000.00 | 2500.00 | 1800.00 | 2026-04-30 | Bank Transfer |
| 507 | 7 | 65000.00 | 5000.00 | 3000.00 | 2026-04-30 | Check |
| 508 | 8 | 47000.00 | 1200.00 | 1100.00 | 2026-04-30 | Bank Transfer |
| 509 | 9 | 54000.00 | 2200.00 | 1600.00 | 2026-04-30 | Bank Transfer |
| 510 | 10 | 44000.00 | 1000.00 | 1000.00 | 2026-04-30 | Bank Transfer |
| 511 | 11 | 63000.00 | 4000.00 | 2500.00 | 2026-04-30 | Check |
| 512 | 12 | 39000.00 | 600.00 | 900.00 | 2026-04-30 | Bank Transfer |
| 513 | 13 | 51000.00 | 2000.00 | 1500.00 | 2026-04-30 | Bank Transfer |
| 514 | 14 | 46000.00 | 1100.00 | 1100.00 | 2026-04-30 | Bank Transfer |
| 515 | 15 | 45500.00 | 1800.00 | 1400.00 | 2026-04-30 | Bank Transfer |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL |

# • INSERTING VALUES INTO TABLES (SALES) :

```
(5001, '2026-04-01', 101, 1001, 1, 1, 350.50),
(5002, '2026-04-02', 102, 1005, 4, 2, 1100.00),
(5003, '2026-04-03', 105, 1010, 10, 5, 2250.00),
(5004, '2026-04-05', 112, 1002, 15, 1, 2999.00),
(5005, '2026-04-07', 121, 1020, 1, 10, 250.00),
(5006, '2026-04-10', 114, 1015, 4, 1, 1999.50),
(5007, '2026-04-12', 108, 1001, 10, 2, 8220.00),
(5008, '2026-04-15', 123, 1030, 15, 1, 29999.99),
(5009, '2026-04-18', 111, 1008, 1, 3, 2399.97),
(5010, '2026-04-20', 103, 1012, 4, 1, 45000.00),
(5011, '2026-04-22', 119, 1004, 10, 2, 700.00),
(5012, '2026-04-25', 125, 1022, 15, 1, 699.00),
(5013, '2026-04-26', 104, 1003, 1, 1, 1199.99),
(5014, '2026-04-27', 106, 1018, 4, 4, 1199.96),
(5015, '2026-04-28', 120, 1009, 10, 1, 499.99);
```

| sale_id | sale_date | product_id | customer_id | employee_id | quantity | total_amount |
|---|---|---|---|---|---|---|
| 5001 | 2026-04-01 | 101 | 1001 | 1 | 1 | 350.50 |
| 5002 | 2026-04-02 | 102 | 1005 | 4 | 2 | 1100.00 |
| 5003 | 2026-04-03 | 105 | 1010 | 10 | 5 | 2250.00 |
| 5004 | 2026-04-05 | 112 | 1002 | 15 | 1 | 2999.00 |
| 5005 | 2026-04-07 | 121 | 1020 | 1 | 10 | 250.00 |
| 5006 | 2026-04-10 | 114 | 1015 | 4 | 1 | 1999.50 |
| 5007 | 2026-04-12 | 108 | 1001 | 10 | 2 | 8220.00 |
| 5008 | 2026-04-15 | 123 | 1030 | 15 | 1 | 29999.99 |
| 5009 | 2026-04-18 | 111 | 1008 | 1 | 3 | 2399.97 |
| 5010 | 2026-04-20 | 103 | 1012 | 4 | 1 | 45000.00 |
| 5011 | 2026-04-22 | 119 | 1004 | 10 | 2 | 700.00 |
| 5012 | 2026-04-25 | 125 | 1022 | 15 | 1 | 699.00 |
| 5013 | 2026-04-26 | 104 | 1003 | 1 | 1 | 1999.99 |
| 5014 | 2026-04-27 | 106 | 1018 | 4 | 4 | 1199.96 |
| 5015 | 2026-04-28 | 120 | 1009 | 10 | 1 | 499.99 |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL |

# • INSERTING VALUES INTO TABLES (CUSTOMERS) :

```
(1001, 'Rahul', 'Sharma', 'rahul.s@email.com', 'Mumbai', 'Gold'),
(1002, 'Anjali', 'Mehta', 'anjali.m@email.com', 'Delhi', 'Silver'),
(1003, 'Vikram', 'Goel', 'vikram.g@email.com', 'Bangalore', 'Bronze'),
(1004, 'Sita', 'Patel', 'sita.p@email.com', 'Mumbai', 'Gold'),
(1005, 'Amit', 'Verma', 'amit.v@email.com', 'Pune', 'Silver'),
(1006, 'Sneha', 'Rao', 'sneha.r@email.com', 'Chennai', 'Bronze'),
(1007, 'Karan', 'Singh', 'karan.s@email.com', 'Delhi', 'Gold'),
(1008, 'Neha', 'Gupta', 'neha.g@email.com', 'Mumbai', 'Silver'),
(1009, 'Rohan', 'Joshi', 'rohan.j@email.com', 'Pune', 'Gold'),
(1010, 'Pooja', 'Nair', 'pooja.n@email.com', 'Bangalore', 'Bronze'),
(1011, 'Arjun', 'Iyer', 'arjun.i@email.com', 'Chennai', 'Silver'),
(1012, 'Deepa', 'Reddy', 'deepa.r@email.com', 'Hyderabad', 'Gold'),
(1013, 'Yash', 'Bansal', 'yash.b@email.com', 'Delhi', 'Bronze'),
(1014, 'Kavita', 'Desai', 'kavita.d@email.com', 'Mumbai', 'Silver'),
(1015, 'Abhishek', 'Mishra', 'abhishek.m@email.com', 'Kolkata', 'Gold'),
(1016, 'Ishani', 'Kapoor', 'ishani.k@email.com', 'Pune', 'Bronze'),
(1017, 'Manish', 'Pandey', 'manish.p@email.com', 'Delhi', 'Silver'),
(1018, 'Riya', 'Saxena', 'riya.s@email.com', 'Bangalore', 'Gold'),
(1019, 'Sandeep', 'Chopra', 'sandeep.c@email.com', 'Mumbai', 'Bronze'),
(1020, 'Megha', 'Kulkarni', 'megha.k@email.com', 'Pune', 'Silver'),
(1021, 'Varun', 'Malhotra', 'varun.m@email.com', 'Delhi', 'Gold'),
(1022, 'Tanya', 'Suri', 'tanya.s@email.com' ,'Hyderabad', 'Bronze'),
(1023, 'Prateek', 'Vats', 'prateek.v@email.com', 'Kolkata', 'Silver'),
(1024, 'Simran', 'Kaur', 'simran.k@email.com', 'Bangalore', 'Gold'),
(1025, 'Gaurav', 'Shah', 'gaurav.s@email.com', 'Mumbai', 'Silver'),
(1026, 'Preeti', 'Tyagi', 'preeti.t@email.com', 'Pune', 'Bronze'),
(1027, 'Nikhil', 'Garg', 'nikhil.g@email.com', 'Delhi', 'Gold'),
(1028, 'Shweta', 'Dutta', 'shweta.d@email.com', 'Kolkata', 'Silver'),
(1029, 'Vivek', 'Bhardwaj', 'vivek.b@email.com', 'Bangalore', 'Bronze'),
(1030, 'Aditi', 'Sarma', 'aditi.s@email.com', 'Chennai', 'Gold');
```

| customer_id | first_name | last_name | email | city | membership_level |
|---|---|---|---|---|---|
| 1001 | Rahul | Sharma | rahul.s@email.com | Mumbai | Gold |
| 1002 | Anjali | Mehta | anjali.m@email.com | Delhi | Silver |
| 1003 | Vikram | Goel | vikram.g@email.com | Bangalore | Bronze |
| 1004 | Sita | Patel | sita.p@email.com | Mumbai | Gold |
| 1005 | Amit | Verma | amit.v@email.com | Pune | Silver |
| 1006 | Sneha | Rao | sneha.r@email.com | Chennai | Bronze |
| 1007 | Karan | Singh | karan.s@email.com | Delhi | Gold |
| 1008 | Neha | Gupta | neha.g@email.com | Mumbai | Silver |
| 1009 | Rohan | Joshi | rohan.j@email.com | Pune | Gold |
| 1010 | Pooja | Nair | pooja.n@email.com | Bangalore | Bronze |
| 1011 | Arjun | Iyer | arjun.i@email.com | Chennai | Silver |
| 1012 | Deepa | Reddy | deepa.r@email.com | Hyderabad | Gold |
| 1013 | Yash | Bansal | yash.b@email.com | Delhi | Bronze |
| 1014 | Kavita | Desai | kavita.d@email.com | Mumbai | Silver |
| 1015 | Abhishek | Mishra | abhishek.m@email.... | Kolkata | Gold |
| 1016 | Ishani | Kapoor | ishani.k@email.com | Pune | Bronze |
| 1017 | Manish | Pandey | manish.p@email.com | Delhi | Silver |
| 1018 | Riya | Saxena | riya.s@email.com | Bangalore | Gold |
| 1019 | Sandeep | Chopra | sandeep.c@email.com | Mumbai | Bronze |
| 1020 | Megha | Kulkarni | megha.k@email.com | Pune | Silver |
| 1021 | Varun | Malhotra | varun.m@email.com | Delhi | Gold |
| 1022 | Tanya | Suri | tanya.s@email.com | Hyderabad | Bronze |
| 1023 | Prateek | Vats | prateek.v@email.com | Kolkata | Silver |
| 1024 | Simran | Kaur | simran.k@email.com | Bangalore | Gold |
| 1025 | Gaurav | Shah | gaurav.s@email.com | Mumbai | Silver |
| 1026 | Preeti | Tyagi | preeti.t@email.com | Pune | Bronze |
| 1027 | Nikhil | Garg | nikhil.g@email.com | Delhi | Gold |
| 1028 | Shweta | Dutta | shweta.d@email.com | Kolkata | Silver |
| 1029 | Vivek | Bhardwaj | vivek.b@email.com | Bangalore | Bronze |
| 1030 | Aditi | Sarma | aditi.s@email.com | Chennai | Gold |
| NULL | NULL | NULL | NULL | NULL | NULL |

# (BASIC RETRIEVAL & FILTERING)

Q

1) Retrieve all customers who hold a 'Gold' membership.

A

```
SELECT first_name, last_name, email, membership_level
FROM customers
WHERE membership_level = 'Gold';
```

| first_name | last_name | email | membership_level |
|---|---|---|---|
| Rahul | Sharma | rahul.s@email.com | Gold |
| Sita | Patel | sita.p@email.com | Gold |
| Karan | Singh | karan.s@email.com | Gold |
| Rohan | Joshi | rohan.j@email.com | Gold |
| Deepa | Reddy | deepa.r@email.com | Gold |
| Abhishek | Mishra | abhishek.m@email.com | Gold |
| Riya | Saxena | riya.s@email.com | Gold |
| Varun | Malhotra | varun.m@email.com | Gold |
| Simran | Kaur | simran.k@email.com | Gold |
| Nikhil | Garg | nikhil.g@email.com | Gold |
| Aditi | Sarma | aditi.s@email.com | Gold |

# (BASIC RETRIEVAL & FILTERING)

Q

**2) List all products that belong to the 'Audio' category ?**

A

```
SELECT * FROM products
WHERE category =
'Audio';
```

| product_id | product_name | price | category | stock_quantity | rating |
|---|---|---|---|---|---|
| 104 | Noise Cancelling Headphones | 1999.99 | Audio | 30 | 4.70 |
| 111 | Bluetooth Speaker V4 | 799.99 | Audio | 75 | 4.20 |
| 117 | Studio Microphone | 549.00 | Audio | 33 | 4.80 |
| 122 | Wireless Lapel Mic | 899.00 | Audio | 55 | 4.60 |
| NULL | NULL | NULL | NULL | NULL | NULL |

# (BASIC RETRIEVAL & FILTERING)

**3) Find the names of all customers living in 'Mumbai'?**

```
SELECT first_name, last_name FROM customers WHERE city = 'Mumbai';
```

| | first_name | last_name |
|---|---|---|
| ▶ | Rahul | Sharma |
| | Sita | Patel |
| | Neha | Gupta |
| | Kavita | Desai |
| | Sandeep | Chopra |
| | Gaurav | Shah |

# (BASIC RETRIEVAL & FILTERING)

**4) Show the unique list of membership levels available at JD_Ltd ?**

```
SELECT DISTINCT
membership_level
FROM customers;
```

| | membership_level |
|---|---|
| ▶ | Gold |
| | Silver |
| | Bronze |

# (BASIC RETRIEVAL & FILTERING)

**5) List all suppliers with a lead time of less than 5 days ?**

```
SELECT supplier_name
FROM suppliers WHERE
lead_time_days < 5;
```

| supplier_name |
| --- |
| Bright Light Co. |
| Eco-Packaging Inc |

# (AGGREGATIONS FUNCTIONS)

**1) Which city has the highest number of customers ?**

```
SELECT city, COUNT(*)
as count
FROM customers
GROUP BY city
ORDER BY count DESC
LIMIT 1;
```

| | city | count |
|---|---|---|
| ▶ | Mumbai | 6 |

# (AGGREGATIONS FUNCTIONS)

**2) Calculate the total revenue generated by all sales combined ?**

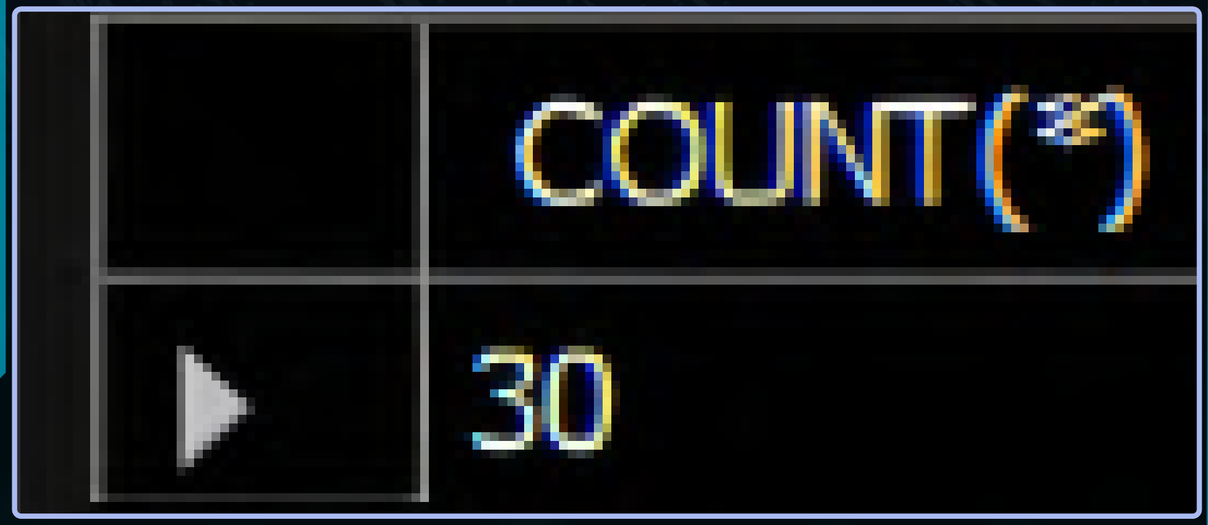
```
SELECT
SUM(total_amount)
FROM sales;
```

| SUM(total_amount) |
| --- |
| 99667.90 |

# (AGGREGATIONS FUNCTIONS)

**3) How many total customers are registered in the customers table?**

```
SELECT COUNT(*) FROM
customers;
```



# (AGGREGATIONS FUNCTIONS)

Q

**4) What is the price of the most expensive product in the store ?**

A

```
SELECT product_name,
price AS highest_price
FROM products
WHERE price = (SELECT
MAX(price) FROM
products);
```

| | product_name | highest_price |
|---|---|---|
| ▶ | Ultra-Wide Monitor 34 | 45000.00 |

# (AGGREGATIONS/JOINS FUNCTIONS)

Q

5) What is the average price of a 'Ergonomic Mouse M2' product which sold at JD_Ltd ?

A

```
SELECT p.product_name,
AVG(s.total_amount /
s.quantity) AS
avg_sold_price
FROM sales s
JOIN products p ON
s.product_id = p.product_id
WHERE p.product_name =
'Ergonomic Mouse M2'
GROUP BY p.product_name;
```

| product_name | avg_sold_price |
|---|---|
| Ergonomic Mouse M2 | 550.0000000000 |

# (JOINS AND SUBQUERIES)

Q

1) Which employee has generated the most revenue for the company ?

A

```
SELECT e.first_name,
SUM(s.total_amount) as
revenue
FROM employees e
JOIN sales s ON
e.employee_id =
s.employee_id
GROUP BY e.employee_id
ORDER BY revenue DESC
LIMIT 1;
```

| first_name | revenue |
|---|---|
| Priya | 49299.46 |

# (JOINS AND SUBQUERIES)

**2) Show the total sales amount for each city. (Requires joining sales and customers) ?**

```
SELECT c.city,
SUM(s.total_amount)
as total_sales
FROM sales s
JOIN customers c ON
s.customer_id =
c.customer_id
GROUP BY c.city;
```

| | city | total_sales |
|---|---|---|
| ▶ | Mumbai | 11670.47 |
| | Pune | 1849.99 |
| | Bangalore | 5449.95 |
| | Delhi | 2999.00 |
| | Kolkata | 1999.50 |
| | Chennai | 29999.99 |
| | Hyderabad | 45699.00 |

# (JOINS AND SUBQUERIES)

**3) List all products that have never been sold ?**

```
SELECT product_name
FROM products
WHERE product_id NOT
IN (SELECT product_id
FROM sales);
```



| product_name |
| --- |
| Smart LED Desk Lamp |
| HD Webcam 1080p |
| Aluminum Laptop Stand |
| Gaming Chair Pro |
| External Hard Drive 2TB |
| Standing Desk Frame |
| Studio Microphone |
| VR Headset Gen 2 |
| Wireless Lapel Mic |
| Anti-Static Work Mat |

# (JOINS AND SUBQUERIES)

**4) Find the supplier who provides the most expensive product in the inventory ?**

```
SELECT su.supplier_name,
p.product_name, p.price
FROM products p
JOIN suppliers su ON
p.category =
su.category_specialty
ORDER BY p.price DESC
LIMIT 1;
```

| supplier_name | product_name | price |
|---|---|---|
| Visionary Displays | Ultra-Wide Monitor 34 | 45000.00 |

# (JOINS AND SUBQUERIES)

**5) We need to see every single product from the products table, even if it has never been sold yet ?**

```
SELECT
p.product_id,
p.product_name,
p.category, s.sale_id,
s.total_amount
FROM products p
LEFT JOIN sales s ON
p.product_id =
s.product_id;
```

| product_id | product_name | category | sale_id | total_amount |
|---|---|---|---|---|
| 101 | Mechanical Keyboard K9 | Peripherals | 5001 | 350.50 |
| 102 | Ergonomic Mouse M2 | Peripherals | 5002 | 1100.00 |
| 103 | Ultra-Wide Monitor 34 | Display | 5010 | 45000.00 |
| 104 | Noise Cancelling Headphones | Audio | 5013 | 1999.99 |
| 105 | USB-C Hub 7-in-1 | Accessories | 5003 | 2250.00 |
| 106 | Wireless Charging Pad | Accessories | 5014 | 1199.96 |
| 107 | Smart LED Desk Lamp | Office | NULL | NULL |
| 108 | Portable SSD 1TB | Storage | 5007 | 8220.00 |
| 109 | HD Webcam 1080p | Audio/Video | NULL | NULL |
| 110 | Aluminum Laptop Stand | Office | NULL | NULL |
| 111 | Bluetooth Speaker V4 | Audio | 5009 | 2399.97 |
| 112 | 4K Mirrorless Camera | Photography | 5004 | 2999.00 |
| 113 | Gaming Chair Pro | Furniture | NULL | NULL |
| 114 | Smartwatch Series 5 | Wearables | 5006 | 1999.50 |
| 115 | External Hard Drive 2TB | Storage | NULL | NULL |
| 116 | Standing Desk Frame | Furniture | NULL | NULL |
| 117 | Studio Microphone | Audio | NULL | NULL |
| 118 | VR Headset Gen 2 | Gaming | NULL | NULL |
| 119 | Mechanical Keycaps Set | Accessories | 5011 | 700.00 |
| 120 | Vertical Laptop Dock | Office | 5015 | 499.99 |
| 121 | Thunderbolt 4 Cable | Accessories | 5005 | 250.00 |
| 122 | Wireless Lapel Mic | Audio | NULL | NULL |
| 123 | Curved Gaming Monitor | Display | 5008 | 29999.99 |
| 124 | Anti-Static Work Mat | Tools | NULL | NULL |
| 125 | Precision Tool Kit | Tools | 5012 | 699.00 |

# CONCLUSION

The primary motivation behind this project was to completely eliminate data blind spots and engineer a secure, highly scalable relational database backend for **JD_ltd.** Traditional retail and tech infrastructure supply chains often suffer from severe procurement bottlenecks, processing backorders, and unexpected inventory stockouts due to disconnected data layers.
Therefore, this project was driven by the need to:

- **Synchronize Real-**Time Pipelines: Maintain instant data consistency between master product catalogs and live customer transaction ledgers.
- **Simplify Bulk Procurement:** Enable corporate clients and development labs to effortlessly execute complex, multi-item bulk orders in a single checkout transaction without system lag.
- **Establish Pricing Transparency:** Prevent checkout price anomalies, ensure predictable infrastructure forecasting, and completely wipe out hidden fees.